Explainability-Guided SegFormer for Forest Cover Segmentation

Using Attention Consistency Supervision

Research Proposal

Semester 5 Undergraduate Research Project — Target: IEEE Conference Publication

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Abstract

Vision Transformer (ViT)-based segmentation models achieve strong accuracy but their internal attention is typically used only for post-hoc visualization, not as a supervised training signal. Prior work applying attention supervision for interpretability has focused on medical imaging and autonomous driving domains, while forest/non-forest segmentation from aerial and satellite imagery still relies on attention purely as an architectural feature-refinement mechanism, with interpretability evaluated qualitatively if at all. This proposal presents an explainability-guided SegFormer-B0 architecture that treats attention as a first-class training target: an Attention Consistency Loss aligns the model's internal attention with the ground-truth forest mask during training, encouraging the network to focus on canopy regions rather than roads, shadows, or built structures. Because SegFormer's encoder uses spatial-reduction attention rather than standard multi-head self-attention, this work adapts Gradient-weighted Attention Rollout to the SegFormer architecture. A new quantitative metric, Average Attention-Mask Overlap (AAMO), is introduced to evaluate interpretability numerically rather than only visually. The proposed method is benchmarked against a U-Net baseline and a vanilla SegFormer-B0 baseline on a dataset of 5,000 256×256 aerial RGB images with binary forest/non-forest masks, using Dice, IoU, F1, and AAMO as evaluation criteria.

1. Introduction

1.1 Background and Motivation

Semantic segmentation of forest cover from aerial and satellite imagery underpins ecological monitoring, deforestation tracking, and land-use management. Convolutional architectures such as U-Net have long dominated this task, and more recently transformer-based models (SegFormer, Swin-UNet, MobileViT) have been applied due to their ability to capture long-range spatial dependencies. However, the great majority of transformer segmentation work — including recent forest-specific studies — uses attention purely as an architectural mechanism to improve accuracy, and evaluates explainability, when it appears at all, only through qualitative heatmap inspection after training.

This project builds on a supervisor-suggested direction: use an existing explainable AI (XAI) method not merely to visualize what a trained model has learned, but as part of the training process itself — turning attention into a supervised training signal rather than a post-hoc artifact.

1.2 Problem Statement

Transformer attention in segmentation models is not explicitly guided during training and frequently attends to task-irrelevant regions such as roads, shadows, and buildings rather than the class of interest. This reduces both segmentation accuracy at ambiguous boundaries and the trustworthiness of the model's explanations. No existing work addresses this specifically for forest/non-forest segmentation, nor evaluates attention faithfulness quantitatively in this domain.

1.3 Research Objectives

- Design an Attention Consistency Loss that supervises SegFormer-B0's internal attention against the ground-truth forest mask during training.
- Adapt Gradient-weighted Attention Rollout to SegFormer's spatial-reduction attention mechanism.
- Define and evaluate a quantitative interpretability metric (AAMO) alongside standard segmentation metrics.
- Demonstrate that explainability-guided training improves both segmentation accuracy and attention faithfulness relative to a vanilla SegFormer-B0 baseline and a U-Net baseline.

2. Related Work

Existing work relevant to this proposal falls into two distinct families, which is an important distinction for positioning the contribution of this project.

2.1 Attention-as-Architecture: Attention Used to Improve Accuracy

These works embed attention mechanisms (channel, spatial, or gated attention) inside the network architecture purely to improve feature selection and segmentation accuracy. Attention maps are not supervised and are not evaluated as explanations.

Work	Approach	Limitation relative to this proposal
ForResANeXt (2026)	Lightweight embedded attention in residual blocks; attention-gated skip connections; Focal Dice loss	Attention used purely as a feature-refinement mechanism, not evaluated as an explainability signal
Attention-Refined PP-LiteSeg (2026)	Multi-branch channel/spatial/pixel attention fusion for cross-regional UAV forest segmentation	Improves accuracy and generalization only; no interpretability supervision or evaluation
Attention U-Net for deforestation mapping	Standard attention gates in U-Net for forest/deforestation segmentation on Sentinel-2 imagery	Attention gate improves feature selection; not used or measured as an explanation

2.2 Attention-as-Explainability: Attention Used as a Supervised Signal

These works treat attention as an interpretability output and explicitly supervise it during training, but in domains other than remote sensing / forest segmentation, and without a quantitative attention-mask overlap metric.

Work	Approach	Limitation relative to this proposal
eX-ViT (weakly-supervised	Attribute-guided attention loss to	Domain is natural-image weak

Work	Approach	Limitation relative to this proposal
segmentation)	regularize ViT attention maps for interpretability	supervision, not remote sensing; no forest-specific evaluation
TransAttUnet (medical segmentation)	Multi-level guided attention integrated into a U-Net/Transformer hybrid	Applied to medical imaging; does not address remote-sensing class-imbalance or canopy/road confusion
Guiding Attention in End-to-End Driving Models	Explicit attention loss during training to align attention with task-relevant regions	Domain is autonomous driving perception; no segmentation mask supervision or AAMO-style metric

2.3 Research Gap

No identified prior work combines (a) attention supervision as a training signal, (b) the forest/non-forest remote-sensing domain, and (c) a quantitative interpretability metric evaluated alongside standard segmentation metrics. This proposal addresses that combined gap.

3. Proposed Methodology

3.1 Architecture Overview

The proposed pipeline extends SegFormer-B0 with an explainability supervision path applied during training only; at inference, the model runs as a standard SegFormer-B0 with negligible overhead.

- Input RGB image (256×256×3) → Patch Embedding → SegFormer Encoder (4 stages)
- Encoder features → MLP Segmentation Decoder → Predicted Mask
- Encoder attention (from later stages) → Adapted Grad-Rollout → Attention Map
- Ground-truth mask → Gaussian smoothing → Soft Attention Target
- Attention Map vs. Soft Attention Target → Attention Consistency Loss → combined with segmentation loss for backpropagation

3.2 Handling SegFormer's Spatial-Reduction Attention

Standard Attention Rollout assumes full multi-head self-attention across all tokens, as in vanilla ViT. SegFormer's encoder instead uses Efficient Self-Attention with a spatial-reduction ratio on keys and values at earlier stages, which reduces sequence length and changes what an attention weight represents. This proposal adapts Gradient-weighted Attention Rollout to account for the spatial-reduction step, and restricts full-resolution attention aggregation to the later encoder stages where the reduction ratio is smallest or absent, so that resulting attention maps remain spatially meaningful relative to the 256×256 input.

3.3 Mathematical Formulation

Let the input image be $X \in R^{(256\times256\times3)}$ with ground-truth binary mask $Y \in \{0,1\}^{(256\times256)}$.

$$F = E(X),\ \ F = \{F1, F2, F3, F4\}\quad \text{(SegFormer encoder features)}$$
$$P = D(F),\ \ P \in [0,1]^{(256\times256)}\quad \text{(segmentation decoder output)}$$
$$A = \mathrm{Rollout}(F),\ \ A \in [0,1]^{(256\times256)}\quad \text{(adapted attention map)}$$
$$A^* = \mathrm{Gaussian}(Y)\quad \text{(soft attention target)}$$

Loss terms:

$$L\_dice = 1 - (2|P \cap Y|) / (|P| + |Y|)$$
$$L\_bce = -\Sigma\, [Y \log P + (1-Y) \log(1-P)]$$
$$L\_att = \mathrm{MSE}(A, A^*)\quad \text{or}\quad \mathrm{KL}(A \,\|\, A^*)$$

Total training objective:

$$L = L\_dice + \lambda 1 \cdot L\_bce + \lambda 2 \cdot L\_att$$

with initial values $\lambda1 = 1$, $\lambda2 = 0.3$, to be tuned experimentally via validation-set performance.

3.4 Stretch Goal: Boundary Refinement

If the core contribution (Sections 3.1–3.3) is validated with time remaining, a Boundary Refinement Module will be added: boundaries are extracted from both prediction and ground truth via morphological gradient, and a boundary Dice loss L_boundary is added to the total objective, since most forest segmentation errors occur near canopy edges. This is treated as an optional extension, not a core deliverable, to keep the primary contribution well-tested within the semester timeline.

4. Experimental Design

4.1 Dataset

- 5,000 RGB aerial images, 256×256 resolution
- Corresponding binary masks: forest vs. non-forest
- Standard train / validation / test split (e.g. 70% / 15% / 15%), stratified to preserve forest-cover ratio across splits

4.2 Baselines

- U-Net — CNN baseline
- SegFormer-B0 (no attention loss) — isolates the effect of the proposed attention supervision

4.3 Ablation Study

Each configuration is trained and evaluated under identical settings (optimizer, learning rate schedule, augmentation) to isolate the contribution of each component.

Model / Configuration	Dice	IoU	F1	AAMO
U-Net (CNN baseline)	-	-	-	n/a
SegFormer-B0 (no attention loss)	-	-	-	measured, not optimized
SegFormer-B0 + Attention Consistency Loss	-	-	-	optimized
SegFormer-B0 + Attention Consistency + Boundary Loss (stretch goal)	-	-	-	optimized

4.4 Evaluation Metrics

- Segmentation quality: Dice coefficient, IoU, Precision, Recall, F1-score
- Efficiency: parameter count, FLOPs, inference FPS
- Interpretability: Average Attention-Mask Overlap (AAMO) — the normalized overlap between the thresholded attention map and the ground-truth forest mask, reported per model to quantify explanation faithfulness rather than relying on qualitative heatmap inspection alone

5. Expected Contributions

- An Attention Consistency Loss that supervises transformer attention using the segmentation ground truth, applied for the first time (to the best of current literature review) in the forest/non-forest remote-sensing domain.
- An adaptation of Gradient-weighted Attention Rollout to SegFormer's spatial-reduction attention mechanism.
- A quantitative interpretability metric (AAMO) usable alongside standard segmentation metrics in future forest-segmentation studies.
- An empirical comparison showing whether explainability-guided training improves both accuracy and attention faithfulness relative to accuracy-only baselines.

6. Project Plan: Two-Phase Approach

The project is split into two phases to accommodate a short-paper submission at the end of Week 4, ahead of the full 12-week research timeline. Phase 1 targets a work-in-progress short paper reporting the motivation, proposed methodology, and preliminary baseline results. Phase 2 covers implementation and validation of the core contribution (the Attention Consistency Loss and AAMO metric) and produces the full IEEE paper.

6.1 Phase 1 — Short Paper (Weeks 1–4)

6.1.1 Scope

Given the compressed timeline, the Week 4 short paper is scoped as a work-in-progress submission rather than a complete results paper. Persons 1–4 completed their core Weeks 1–2 tasks (dataset pipeline, U-Net baseline, vanilla SegFormer-B0 baseline, evaluation script) ahead of schedule, which created room in Weeks 3–4 to pull part of the Phase 2 contribution forward: a preliminary Attention Consistency Loss training pass and a first-pass AAMO computation. As a result, the short paper can report not only the motivation, proposed architecture, and mathematical formulation, but also an early, small-scale explainability-guided result alongside the U-Net and vanilla SegFormer-B0 baselines — strengthening the submission beyond a baselines-only report, while the full-scale tuning, complete ablation study, and boundary refinement remain Phase 2 work.

6.1.2 Work Distribution

Team member	Weeks	Responsibilities
Person 1 — Data & Baselines	1-2	Completed: cleaned and split the dataset (stratified train/val/test); built the shared augmentation pipeline; trained the U-Net baseline to a working Dice/IoU result.
Person 1 — Data & Baselines	3-4	Extended scope: prepare qualitative dataset and result figures (sample images, masks, predictions) for the paper; run a small augmentation ablation (with vs. without augmentation) as an additional supporting result; support debugging as needed.
Person 2 — Transformer Lead	1-2	Completed: set up the SegFormer-B0 training loop; reproduced a vanilla SegFormer-B0 Dice/IoU baseline.
Person 2 — Transformer Lead	3-4	Extended scope: implement attention-extraction hooks and produce qualitative attention-drift figures (motivation for the paper); additionally run a first integration pass feeding extracted attention into Person 3's early loss implementation to produce a preliminary explainability-guided checkpoint.
Person 3 — Loss & Training	1-2	Completed: wrote the mathematical formulation section; pre-implemented (unit-tested) the Attention Consistency Loss ahead of schedule.
Person 3 — Loss & Training	3-4	Extended scope: integrate the loss with Person 2's attention pipeline; run an early, short-schedule training pass with the Attention Consistency Loss to obtain preliminary numbers for the short paper; begin a first-pass AAMO implementation jointly with Person 4.
Person 4 — Evaluation Lead	1	Completed: built the shared evaluation script (Dice, IoU, F1, parameters, FLOPs).
Person 4 — Evaluation Lead	2-3	Completed: ran evaluation on the U-Net and vanilla SegFormer-B0 checkpoints; produced the first baseline comparison table.
Person 4 — Evaluation Lead	4	Extended scope: evaluate Person 3's preliminary attention-loss checkpoint alongside the baselines; compute a first-pass AAMO

Team member	Weeks	Responsibilities
		number so the short paper can report a rough interpretability result rather than only describing the metric; add an extra baseline (e.g. DeepLabV3+ or Swin-Unet-Tiny) if time allows, for a stronger comparison table.
Person 5 — Writing & Technical Integration Lead	1	Write the full related-work section (two-bucket positioning); set up the shared repository structure and coding conventions so Person 2 and Person 3's code integrates cleanly from the start; implement the Gaussian soft-target generation module (A* = Gaussian(Y)) that will feed the Attention Consistency Loss.
Person 5 — Writing & Technical Integration Lead	2	Draft introduction, problem statement, and proposed-methodology text; produce the architecture/pipeline diagram figure used in Section 3; begin technically overseeing Person 2's attention-extraction work and Person 3's loss implementation, reviewing both against the shared math formulation to keep them consistent with each other; set up the shared experiment-tracking configuration (e.g. Weights & Biases) that all four technical members log to.
Person 5 — Writing & Technical Integration Lead	3	Assemble the first full draft using whatever results exist; directly manage the integration of Person 2's attention output into Person 3's loss code — reviewing the merge, resolving interface mismatches, and running the first joint training pass jointly with them; co-implement the first-pass AAMO metric code alongside Person 4; maintain reproducibility documentation (environment setup, how to run the baselines).
Person 5 — Writing & Technical Integration Lead	4	Finalize the short paper in the IEEE template, manage citations/formatting, and circulate for co-author review; verify Person 3 and Person 4's preliminary attention-loss numbers are consistent and correctly reported before submission; write unit tests for the Attention Consistency Loss and AAMO code to catch integration bugs before Phase 2 scales them up; prepare a short slide deck summarizing the work.

6.1.3 Phase 1 Timeline (Gantt Chart)

Team member	W1	W2	W3	W4
Person 1 — Data & Baselines	C	C	S	S
Person 2 — Transformer Lead	C	C	C	C
Person 3 — Loss & Training	S	S	C	C
Person 4 — Evaluation Lead	C	C	C	C
Person 5 — Writing & Technical Integration Lead	C	C	C	C

Legend: C = core responsibility for that week · S = supporting / lighter involvement

6.2 Phase 2 — Full Paper (Weeks 5–12)

6.2.1 Scope

Phase 2 implements and validates the core contribution: the Attention Consistency Loss, the SegFormer-specific Grad-Rollout adaptation, and the AAMO interpretability metric, followed by the full ablation study and final paper writing. The optional Boundary Refinement Module (Section 3.4) is scheduled as a Week 10 stretch goal, picked up by whichever team member completes their core task early, rather than pre-assigned.

6.2.2 Work Distribution

Team member	Weeks	Responsibilities
Person 1 — Data & Baselines	5-9	Light support role: assist with data-related debugging and any additional augmentation/preprocessing needs as they arise.
Person 2 — Transformer Lead	5-6	Support integration of the attention-extraction pipeline with Person 3's loss implementation.
Person 3 — Loss & Training	5-6	Implement and tune the Attention Consistency Loss (λ sweep); train the full explainability-guided SegFormer-B0 model to convergence.
Person 3 — Loss & Training	7-9	Support the ablation study; help interpret attention-related results.
Person 4 — Evaluation Lead	7-9	Implement the AAMO metric; run the complete ablation study across all model configurations; report results with multiple seeds (mean ± std).
Person 4 — Evaluation Lead	10	Support role: assist whoever picks up the boundary-refinement stretch goal with evaluation.
Open / stretch goal	10	Boundary Refinement Module (optional) — Person 5 is the designated primary owner (having started it in Weeks 7-9); reassigned to whichever other team member has spare capacity if Person 5's integration workload runs long.
Person 5 — Writing & Technical Integration Lead	5-6	Technically oversee Person 3's Attention Consistency Loss tuning — review λ-sweep results, sanity-check training curves and loss behavior; implement the KL-divergence variant of the attention loss as an alternative to MSE so Person 3 can compare both formulations; keep related-work and methodology sections updated in parallel.
Person 5 — Writing & Technical Integration Lead	7-9	Manage the handoff between Person 3's trained model and Person 4's ablation study — confirm checkpoints, configs, and metric definitions (especially AAMO) are used consistently across every ablation row; resolve any discrepancies between training and evaluation code; begin implementing the Boundary Refinement Module (morphological boundary extraction + boundary Dice loss) ahead of the Week 10 stretch-goal slot, using spare capacity from the integration-management workload.
Person 5 — Writing & Technical	10	Primary owner of the Boundary Refinement Module stretch goal: finish integrating it into the training pipeline and hand off the resulting

Team member	Weeks	Responsibilities
Integration Lead		checkpoint to Person 4 for the final ablation row.
Person 5 — Writing & Technical Integration Lead	11-12	Assemble the full paper (results, discussion, conclusion) directly from Person 4's verified ablation table; coordinate final figures; circulate for full co-author review before submission.
All members	11-12	Jointly review and correct the results and discussion sections describing their own contributions.

6.2.3 Phase 2 Timeline (Gantt Chart)

Team member	W5	W6	W7	W8	W9	W10	W11	W12
Person 1 — Data & Baselines	S	S	S	S	S			
Person 2 — Transformer Lead	S	S						
Person 3 — Loss & Training	C	C	S	S	S			
Person 4 — Evaluation Lead			C	C	C	S		
Person 5 — Writing & Technical Integration Lead	S	S	S	S	S	C	C	C

Legend: C = core responsibility for that week · S = supporting / lighter involvement

7. Target Venue

This work targets an IEEE conference appropriate for an undergraduate research contribution in computer vision / remote sensing, such as ICIAfS, ICTer, MERCon, or TENCON, subject to submission deadlines aligned with the project timeline.

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